FALL 2026 | MEE4033

Mechatronics

Mechatronics (MEE4033) · Department of Mechanical Engineering

Lecture 12

Motor Control 1: Sensors for Motor Control

Motor Control (모터제어), 3rd ed., Chapter 10



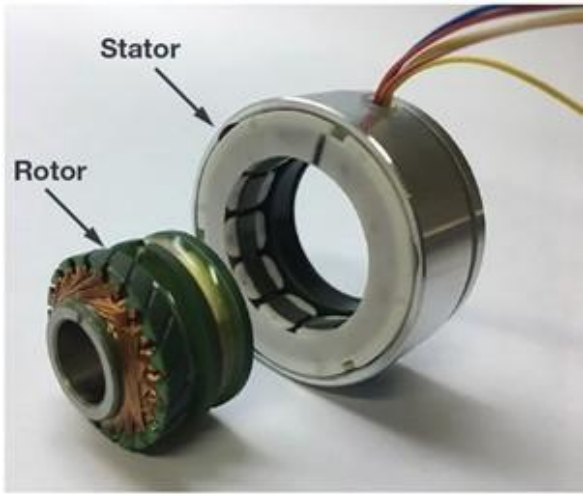
Sogang University · Department of Mechanical Engineering · Prof. Seokhwan Jeong

Sensors for Motor Control

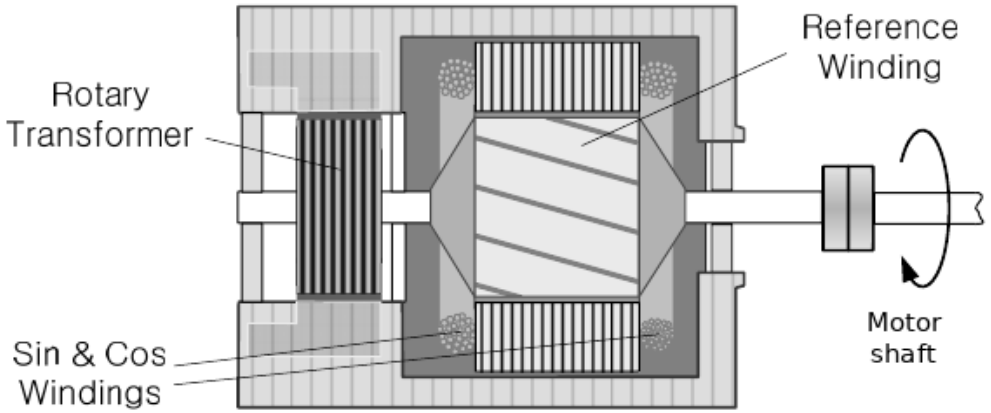
- In a **feedback control system**, a sensor is required to feed back the current state
- For electric motors, sensors provide the **rotational state** of the motor (position, velocity)
- **Analog type**: Synchro, Resolver, Tacho-generator
- **Digital type**: Rotary encoder

Resolver

- **Analog-type** rotary sensor
- Provides absolute position information
- Essentially a **rotary transformer**
- Consists of a **rotor** and a **stator**





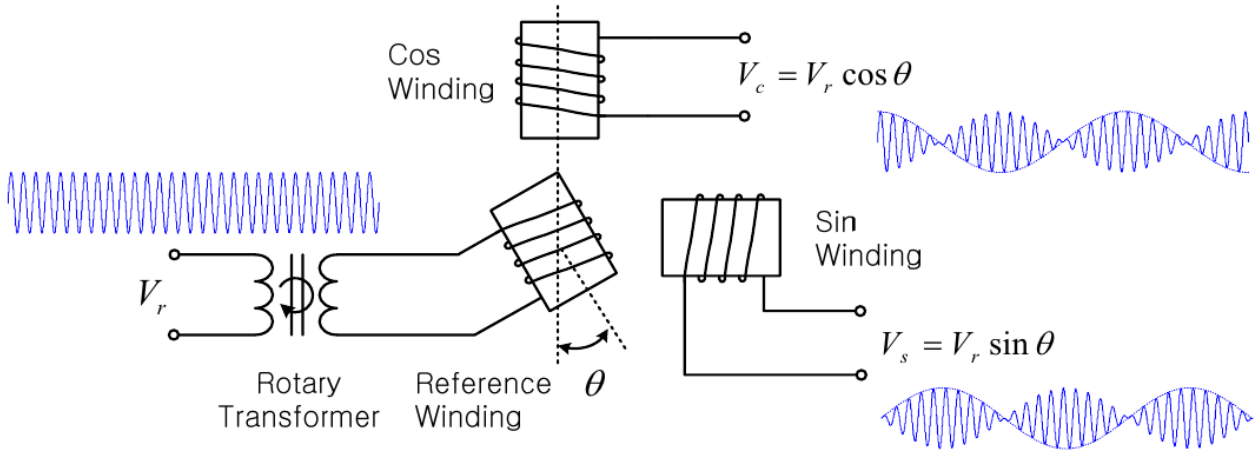


Resolver

Resolver: Windings and Output Signals

- **Primary winding:** the reference winding is excited through the rotary transformer
- **Secondary winding:** the Sin and Cos windings are placed with a **90-degree** phase difference
- The high-frequency signal (**1 to 100 kHz**) in the primary winding is converted to the outputs of the secondary windings





Structure of the windings of the resolver and output signals

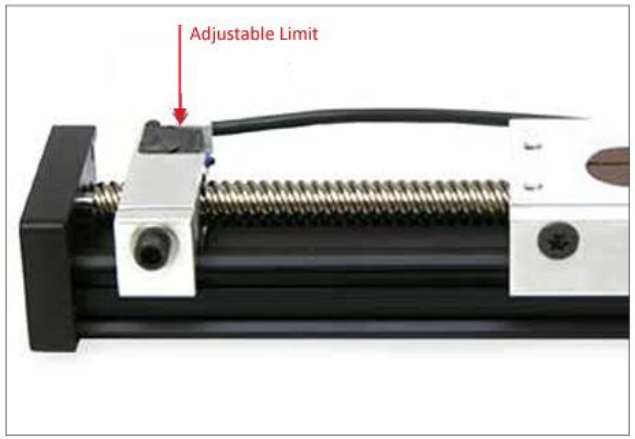
Resolver: Computing the Absolute Angle

- The output amplitudes of the secondary windings are modulated by the **relative rotary position** between the primary and secondary windings
  - 0-degree phase difference: maximum output
  - 90-degree phase difference: zero output
- The **absolute** rotary position can be calculated from the two output signals

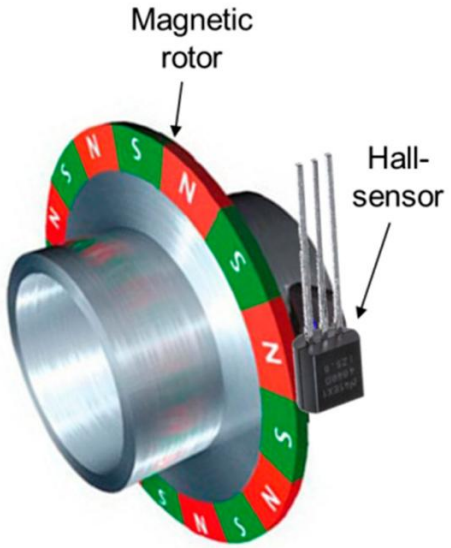
$$\theta = \tan^{-1}\left(\frac{V_c}{V_s}\right) = \tan^{-1}\left(\frac{V_r \sin\theta}{V_r \cos\theta}\right)$$

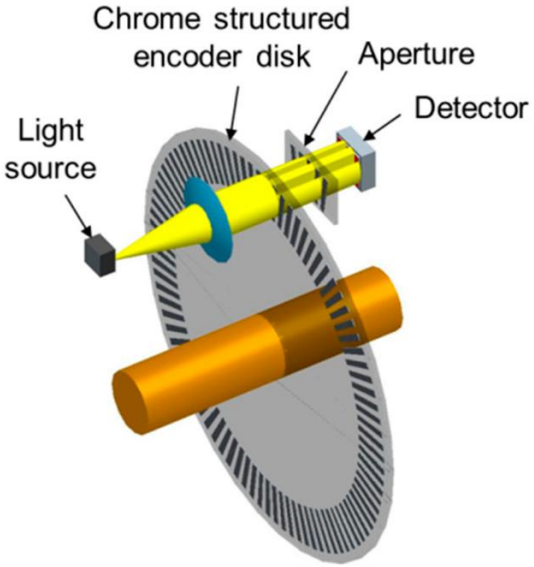
Encoder

- **Digital-type** rotary sensor
- Provides absolute or incremental position information
- Incremental-type encoder: **magnetic** type or **optical** type









Encoder

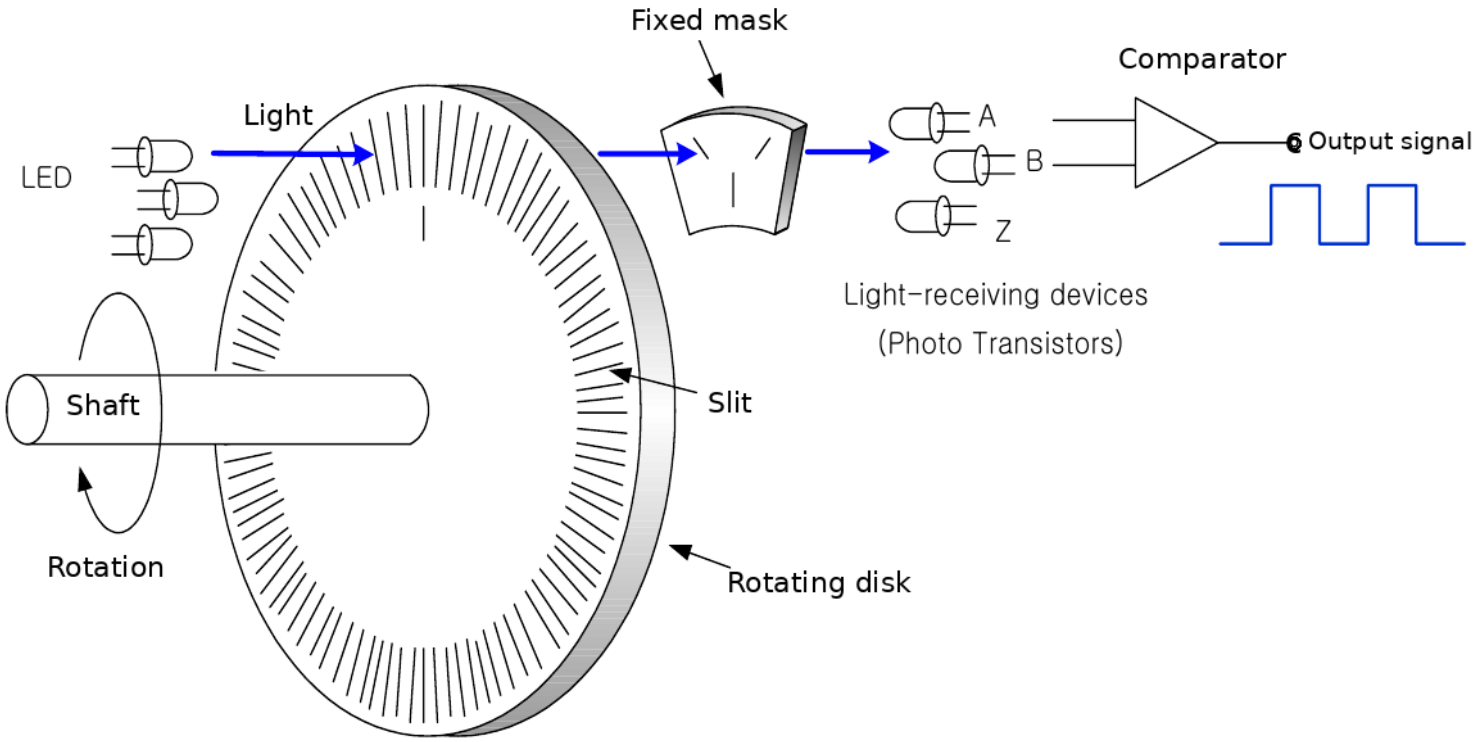
Incremental Encoder

- An incremental encoder outputs rotary information as a **pulse train**
- **Pulses per revolution (PPR)**, e.g. 512, 1024
- Sensitive to shock, vibration, dust and noise
- A **shielded twisted-pair cable** makes the pulse signal robust to noise



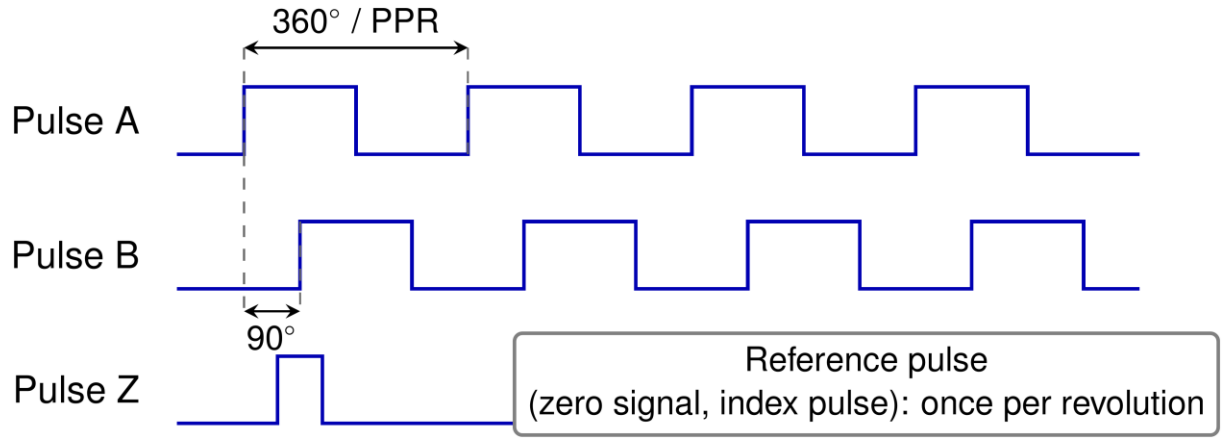
Optical Encoder: Incremental

- The number of slits in the disk is the **PPR**
- Photo transistors convert the light passing through the slits into an electric signal (a **square wave** with 50% duty)



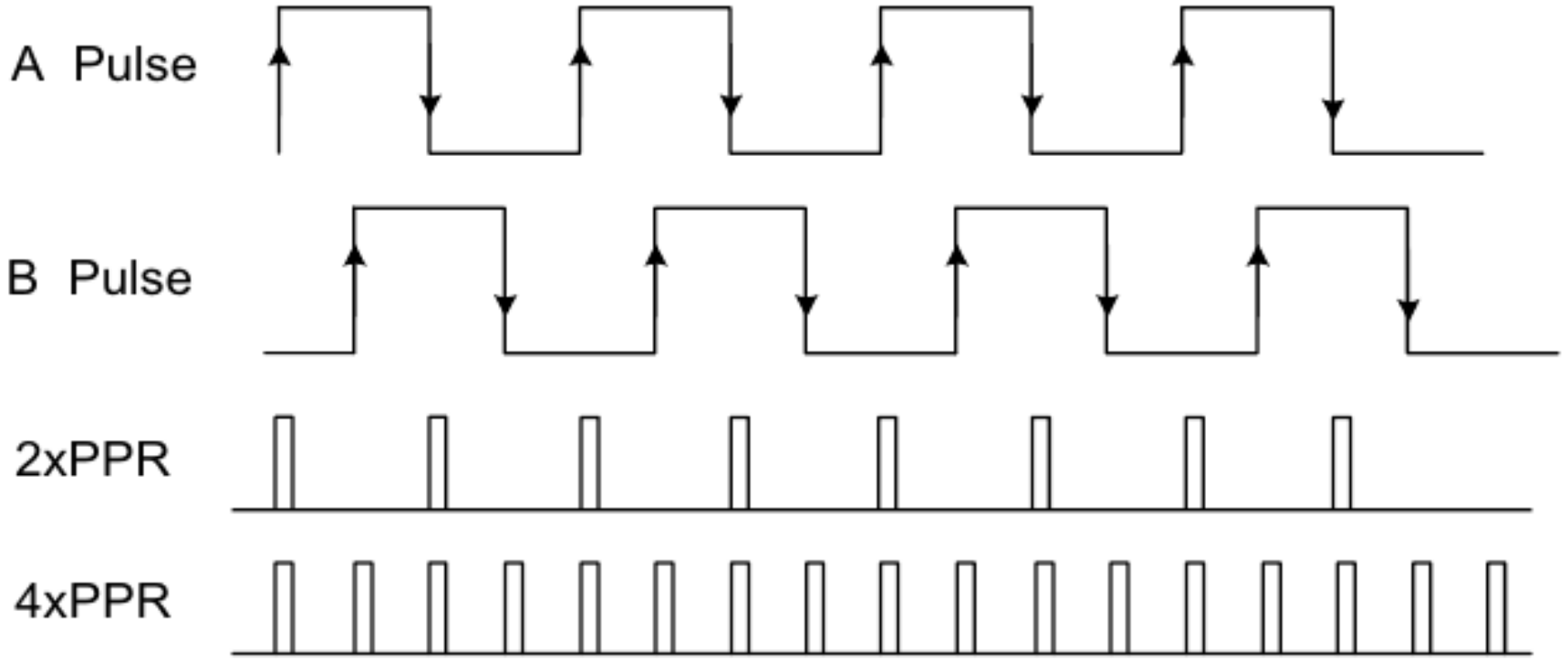
Optical Encoder: Pulse A, Pulse B and Pulse Z

- **Pulse A, Pulse B:**
  - Output PPR pulses per revolution
  - 1 pulse corresponds to 360° / PPR
  - 90-degree phase difference between A and B → the **direction** of rotation can be obtained
- **Pulse Z (reference pulse):**
  - Sets a reference (zero) position once per revolution



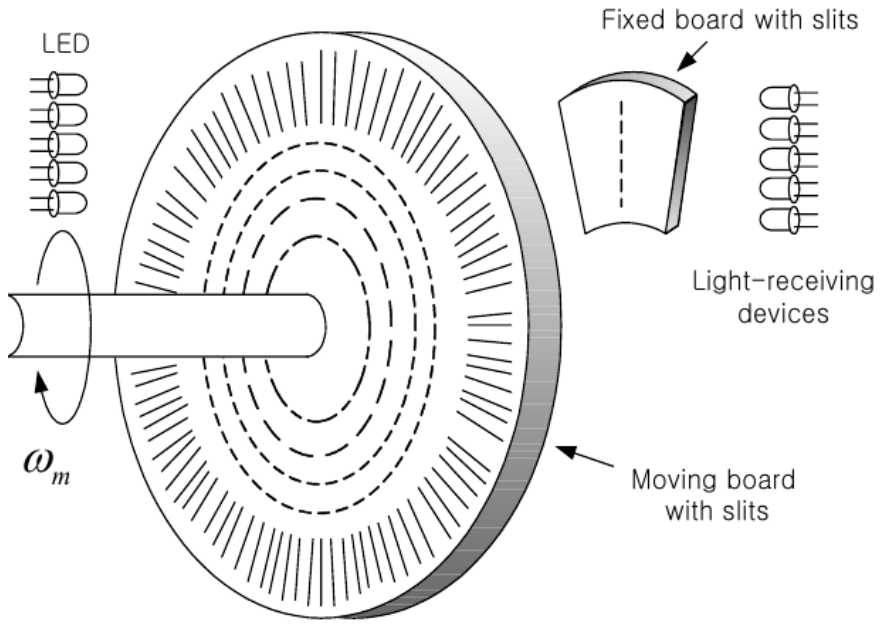
Optical Encoder: 2x and 4x Multiplication

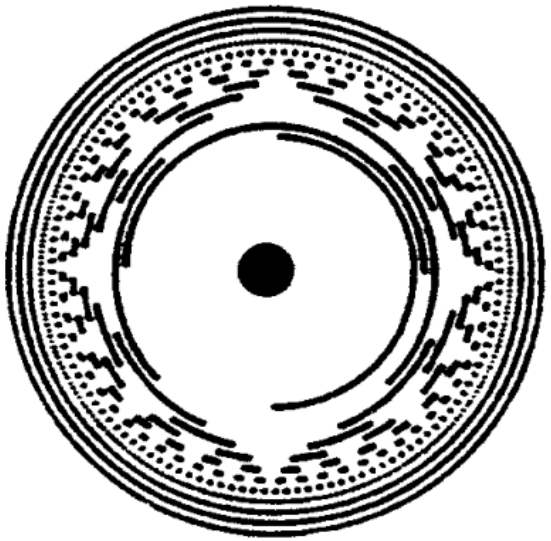
- **2x / 4x multiplication:** counting both edges of A (2x), or both edges of A and B (4x), raises the resolution to 2 PPR or **4 PPR** counts per revolution without changing the disk



Optical Encoder: Absolute

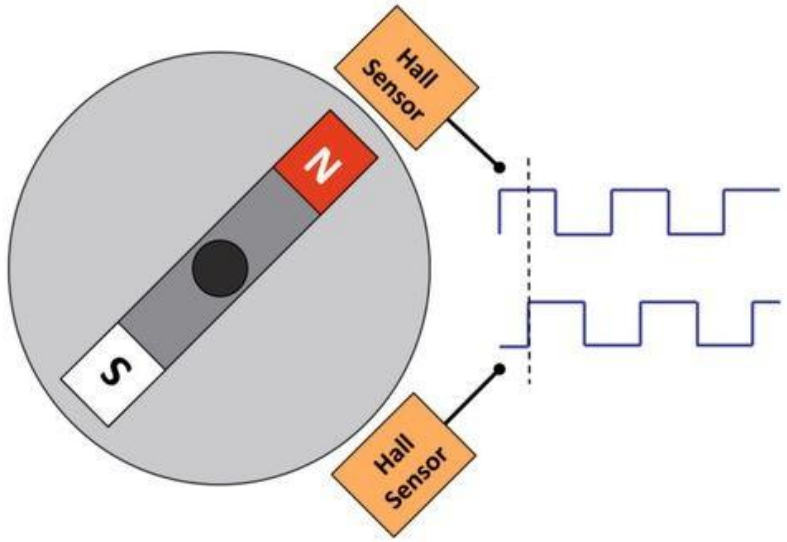
- The digital output pattern depends on the position of the rotor (each angle has a unique code)
- Output codes: binary code, binary-coded decimal, **Gray code**, etc.
- Complex structure, difficult to miniaturize, and high cost

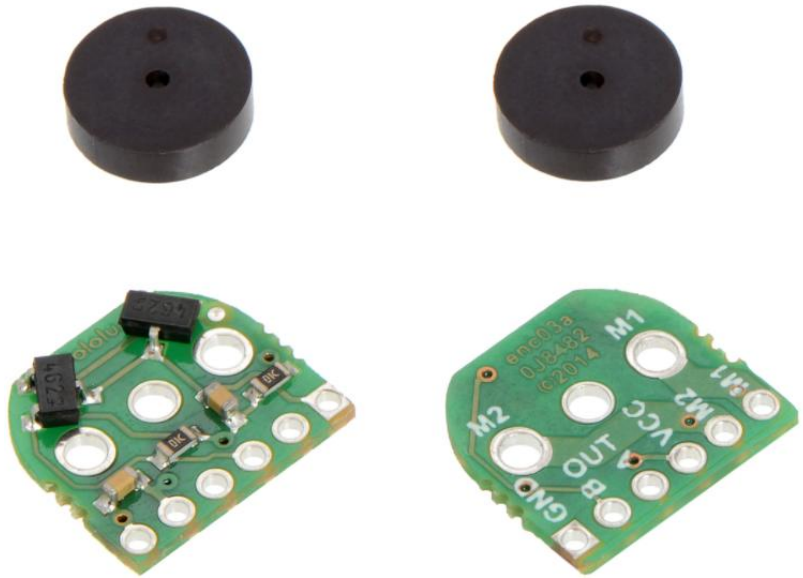




Magnetic Encoder: Incremental

- **Hall effect**: the production of a voltage difference (the Hall voltage) across an electrical conductor, transverse to an electric current in the conductor and to an applied magnetic field perpendicular to the current. Discovered by Edwin Hall in 1879.

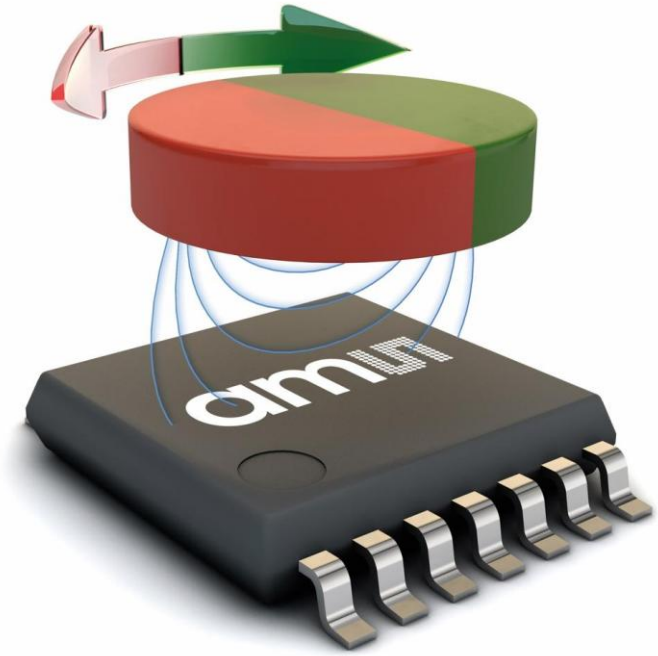




pololu.com/product/2598 (magnetic encoder pair kit for micro metal gearmotors)

Magnetic Encoder: Absolute

- Measures the magnetic field of a diametrically magnetized magnet and calculates the **direction** of the rotor (absolute angle)

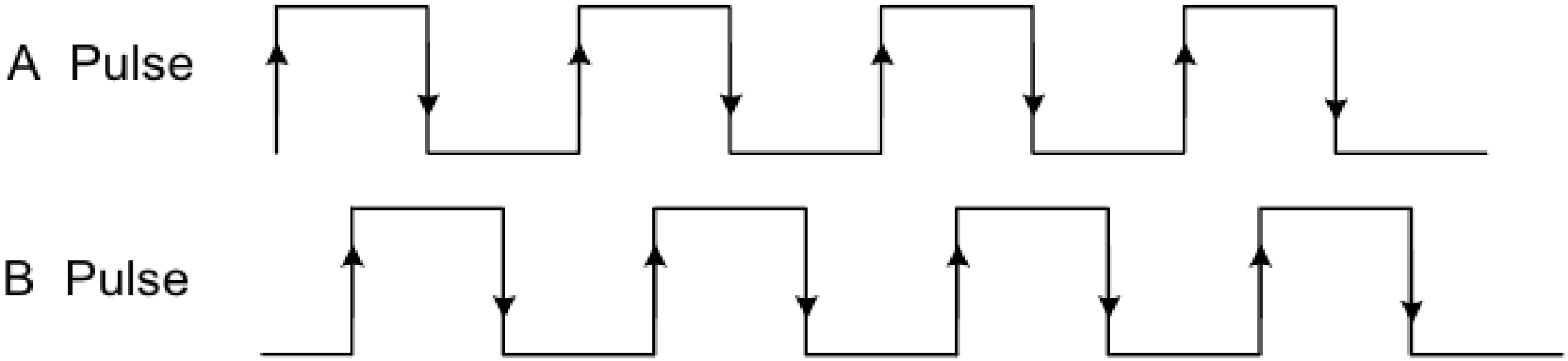




https://www.infineon.com/part/AS5048A

Magnetic Encoder: Incremental (Quadrature Signals)

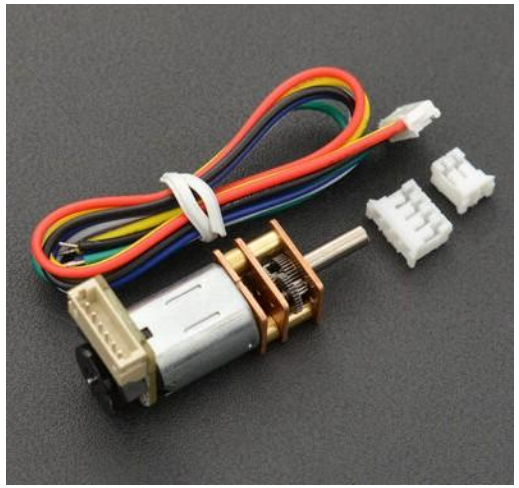
- Two Hall sensor outputs, A and B, 90 degrees apart: which one leads tells the direction, and every edge is a count

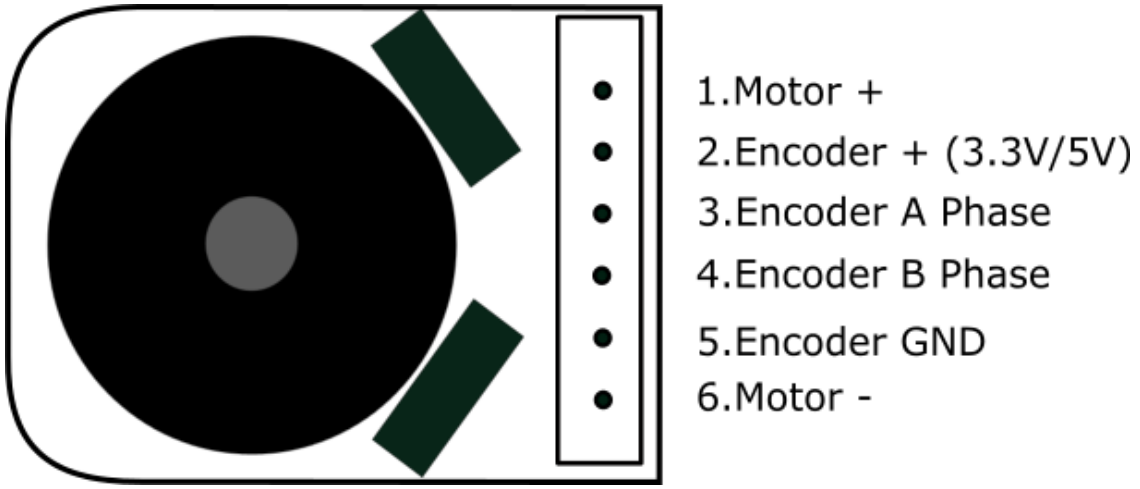


Geared Motor with a Magnetic Incremental Encoder

- Geared motor with a magnetic incremental encoder (kit motor)

dfrobot.com/product-1434.html

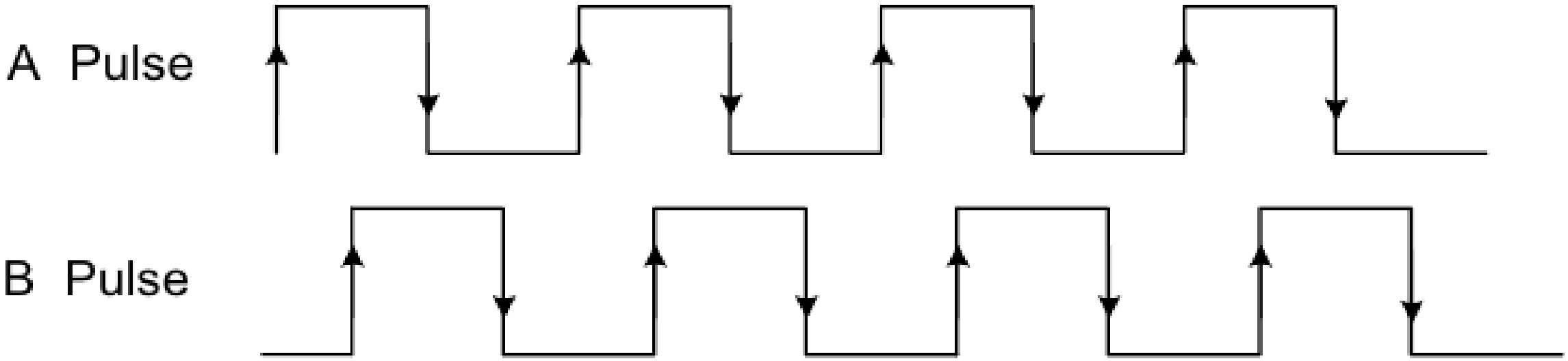




Product description (DFRobot): a micro metal gear motor with a built-in encoder which measures the motor's speed in real time. The average output number of pulses can reach up to 7 x 2 x 150 pulses per revolution. The motor has a long (9.0 mm) D-profile metal output shaft; the brass faceplate has two M1.6 mounting holes. The Hall sensor of the new version has 7 pole pairs, so the encoder resolution is increased 7-fold.

Magnetic Encoder: Incremental (Counting with Interrupts)

- With an external interrupt on **CHANGE** of both A and B, every edge calls the ISR: how many counts per revolution of the encoder disk do we get?



Reading the Encoder with External Interrupts: First Test

```
int interruptPin1 = 2;
int interruptPin2 = 3;

void setup() {
  Serial.begin(115200);

  pinMode(interruptPin1, INPUT_PULLUP);
  pinMode(interruptPin2, INPUT_PULLUP);
  attachInterrupt(digitalPinToInterrupt(interruptPin1), blink1, CHANGE);
  attachInterrupt(digitalPinToInterrupt(interruptPin2), blink2, CHANGE);
}

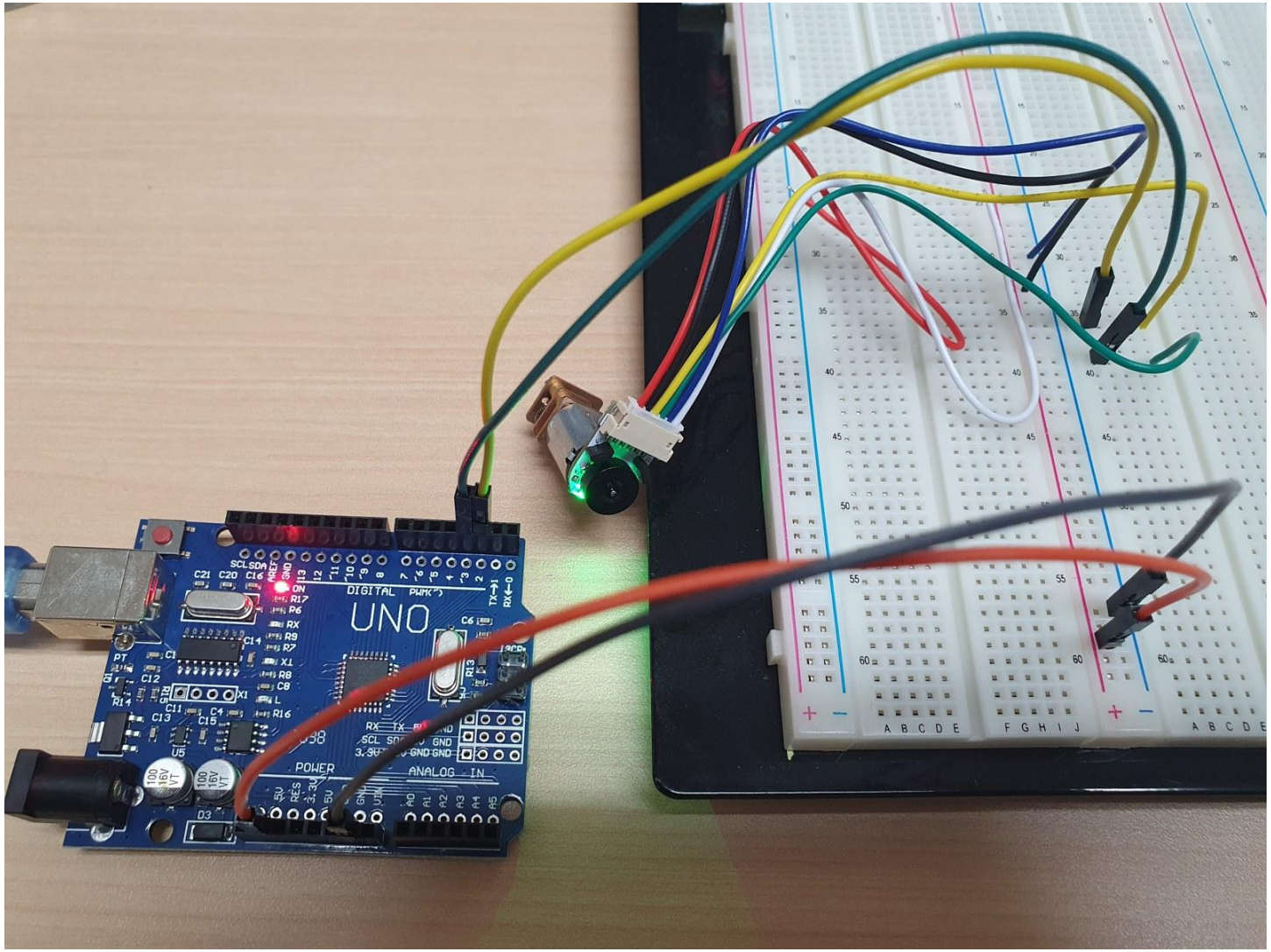
void loop() {
}

void blink1() {
  Serial.println("CHANGE (RISING or FALLING) edge from --- Pin1");
}

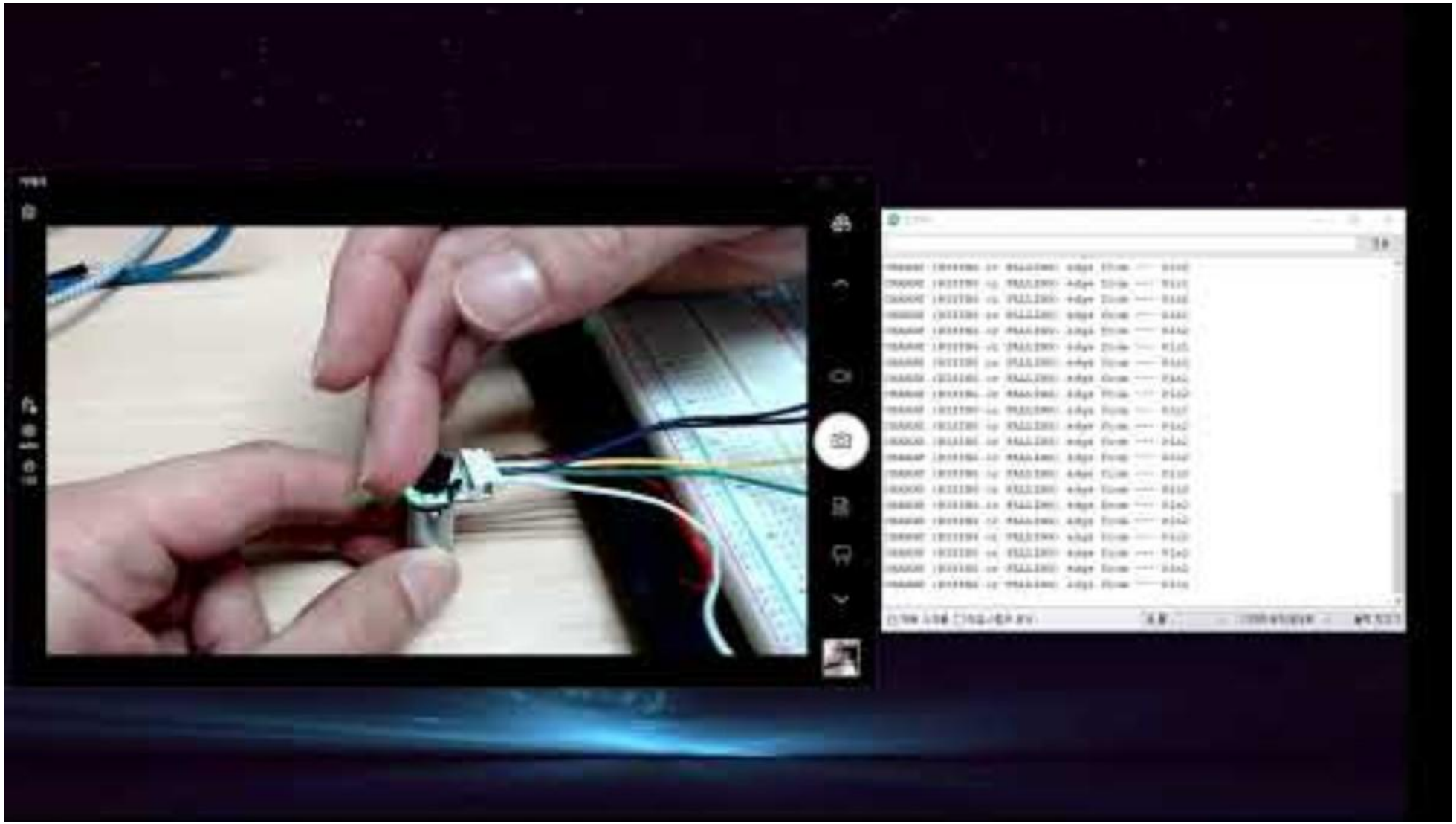
void blink2() {
  Serial.println("CHANGE (RISING or FALLING) edge from --- Pin2");
}
```

- Encoder A and B on pins **2** and **3** (the two external-interrupt pins of the Uno)
- **INPUT PULLUP**: the Hall sensor outputs are open-collector type
- **attachInterrupt(..., CHANGE)**: the ISR runs on both rising and falling edges
- Turn the encoder disk by hand and watch the Serial Monitor (115200 baud)

Wiring: Encoder to Arduino Uno



Demo: Edges on the Serial Monitor



Encoder Counter: Declarations and setup()

```
void loop() {

  if(((int)t%10)<5)
  {
    Ref=10000;
  }
  else
  {
    Ref=0;
  }
 Serial.println(Enc);

}

void Timer_ISR()
{
  t = t + 0.01;

}
```

```
void EncoderCounter() {

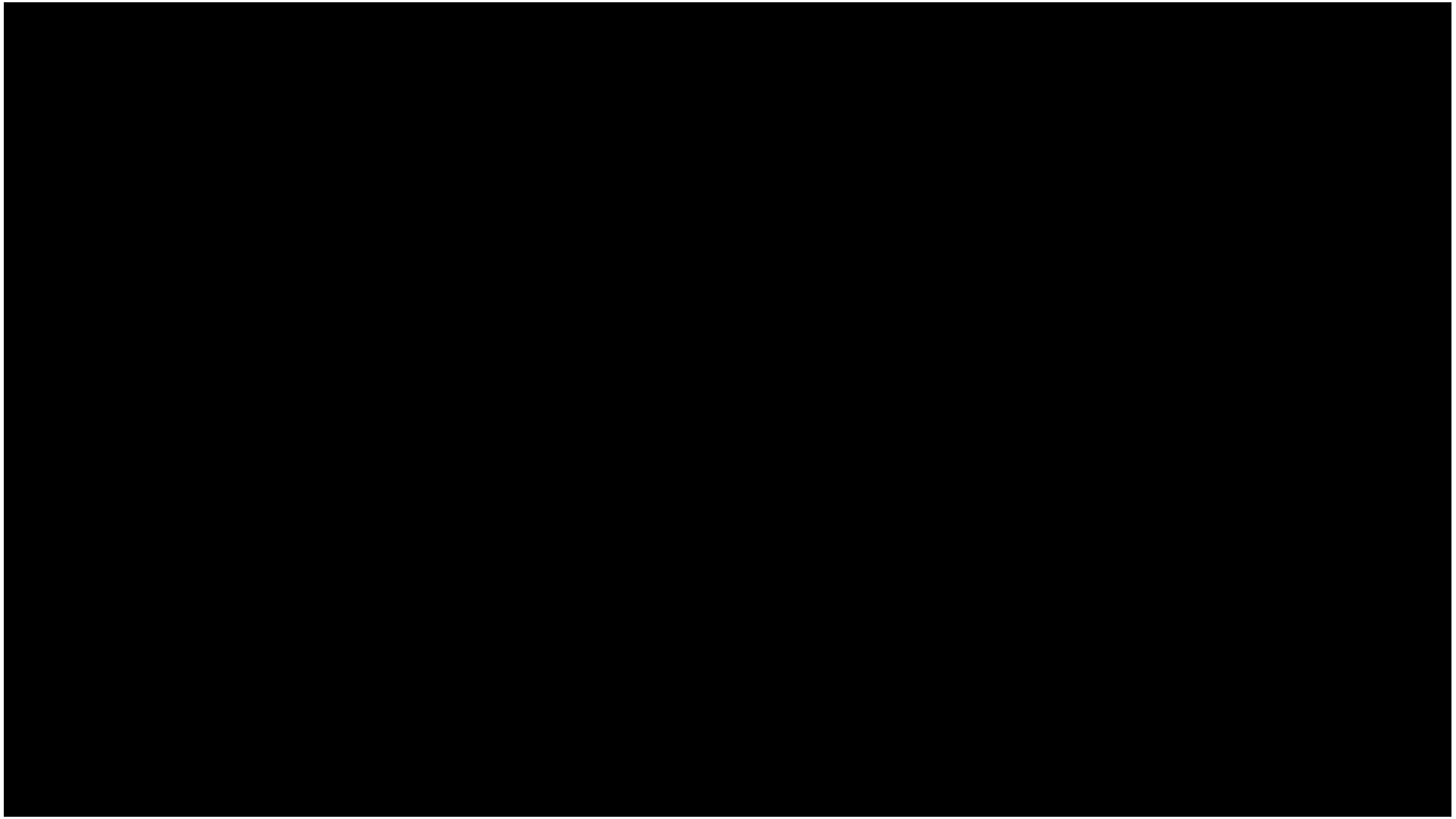
  int HA = digitalRead(interruptPin1);
  int HB = digitalRead(interruptPin2);
  int Dir;

  int HSP = HA + 2 * HB;

  switch (HSP) {
    case 0: if (HSPp == 1) Dir = 1; else if (HSPp == 2) Dir = -1; else Dir = 0; break;
    case 1: if (HSPp == 3) Dir = 1; else if (HSPp == 0) Dir = -1; else Dir = 0; break;
    case 2: if (HSPp == 0) Dir = 1; else if (HSPp == 3) Dir = -1; else Dir = 0; break;
    case 3: if (HSPp == 2) Dir = 1; else if (HSPp == 1) Dir = -1; else Dir = 0; break;
      //default: fault2[no] = 1; break;
  }

  Enc = Enc + Dir;
  HSPp = HSP;
}
```

Demo: Counting One Revolution



Report 4

- Write a program that counts the rotation of the **magnetic incremental encoder** using **external interrupts**, and submit a demo video **(60 pt)**
  - Turn the encoder by hand through one revolution and print the encoder count on the Arduino IDE Serial Monitor
  - Turn the encoder one revolution clockwise, then one revolution counter-clockwise: the count must increase from 0 to 28 and then decrease from 28 to 0. Record this as a video
  - The video must show the encoder and the Serial Monitor window together
  - Your student ID must appear in the video
- Submit: (1) the demo video, (2) the code as code.txt